جامعة مصر للمعلوماتية
EGYPT UNIVERSITY
OF INFORMATICS

Egypt University of Informatics
Computer and Information Systems
Data Analysis Course

The Relationship Between Caffeine Consumption and Sleep Patterns

Submitted by: Farida Osama / 24-101513

Hana Ahmed / 24-101108

Tasneem Ahmed / 24-101166

Abdelrahman Ahmed / 24-101501

10/4/2026

Introduction

Sleep quality is a key factor in human health, affecting concentration, productivity and overall well-being. In recent years, caffeine consumption has increased significantly due to energy drinks, coffee culture and academic pressure.

This study aims to explore the relationship between caffeine intake and sleep behaviour among individuals. Specifically, we investigate whether higher caffeine consumption is associated with reduced sleep duration and poorer sleep quality.

Understanding this relationship is important because caffeine is widely consumed daily, yet its long-term effects are often underestimated.

Research Question

What is the relationship between daily caffeine consumption and sleep duration/quality?

Hypothesis

Higher caffeine consumption is associated with fewer hours of sleep and lower sleep quality.

Population of Interest:

The primary population of interest is university students aged 18-25 in Egypt, as they often have irregular sleep schedules and consume caffeine regularly, However, the survey also included responses from non-students, which add some variation to the data form

Sampling Method:

This study used a convenience sampling method, where responses were collected through an online survey distributed via social media platforms. This approach was chosen due to its efficiency and ability to quickly reach a relatively large number of participants within the available time frame.

However, convenience sampling does not guarantee a fully representative sample of the target population, as participants are self-selected and may share similar characteristics (e.g., age group, social circle, or lifestyle). Despite this limitation, the method was considered appropriate for this study, as it allowed for sufficient data collection to identify general patterns and trends in caffeine consumption and sleep behaviour.

Bias Identification:

Several potential biases were considered in this study. A key limitation is selection bias, as participants were not randomly selected but instead responded voluntarily through an online survey. This may lead to overrepresentation of individuals who are more health-conscious or interested in the topic

Additionally, self-reporting bias may affect accuracy, since participants may not precisely remember or truthfully report their caffeine intake or sleep habits. To minimize bias, the survey questions were kept neutral and did not suggest any expected answers.

Survey Questions:

Are you a university student?

Age?

On average, how many hours do you sleep per night?

How would you rate your sleep quality?

How many caffeinated drinks do you consume per day?

If you consume caffeine, when do you usually consume it?

If you consume caffeine, do you consume it within 4 hours before sleep?

If you consume caffeine, do you think it affects your sleep?

Online survey link: https://docs.google.com/forms/d/e/1FAIpQLScuS7ptcK3ga-AAsSnDcgWtuj2SbwMsCURcGIh2O2Ol27uJeQ/viewform?usp=header

Number of samples collected: 126

Analysis:

Descriptive statistics and visualizations were used to identify patterns and trends in caffeine consumption and sleep behaviour

Figure 1. University Students

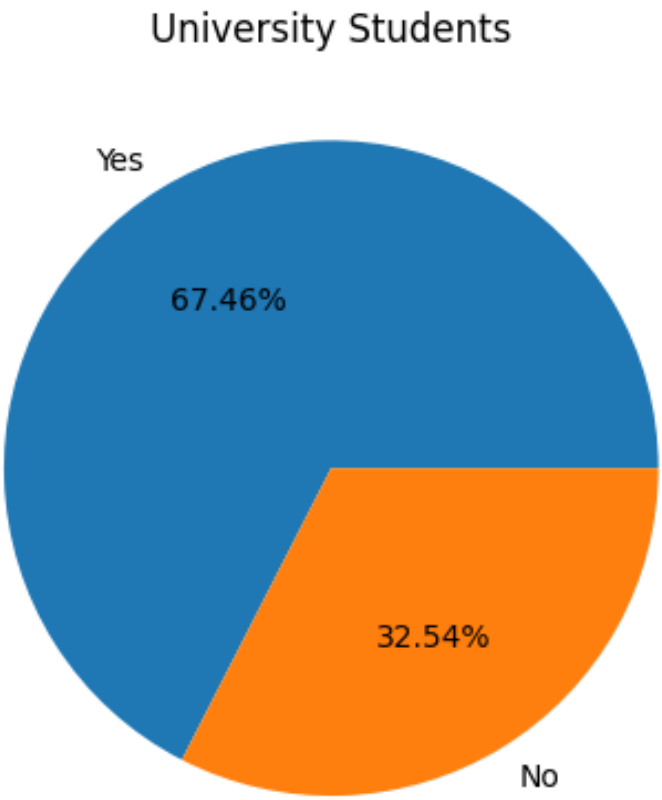

Figure 1 shows the distribution of respondents based on whether they are university students. The majority of respondents are university students, indicating that the sample largely reflects the target population. However, due to the use of convenience sampling, it may not be fully representative.

Figure 2. Age Distribution

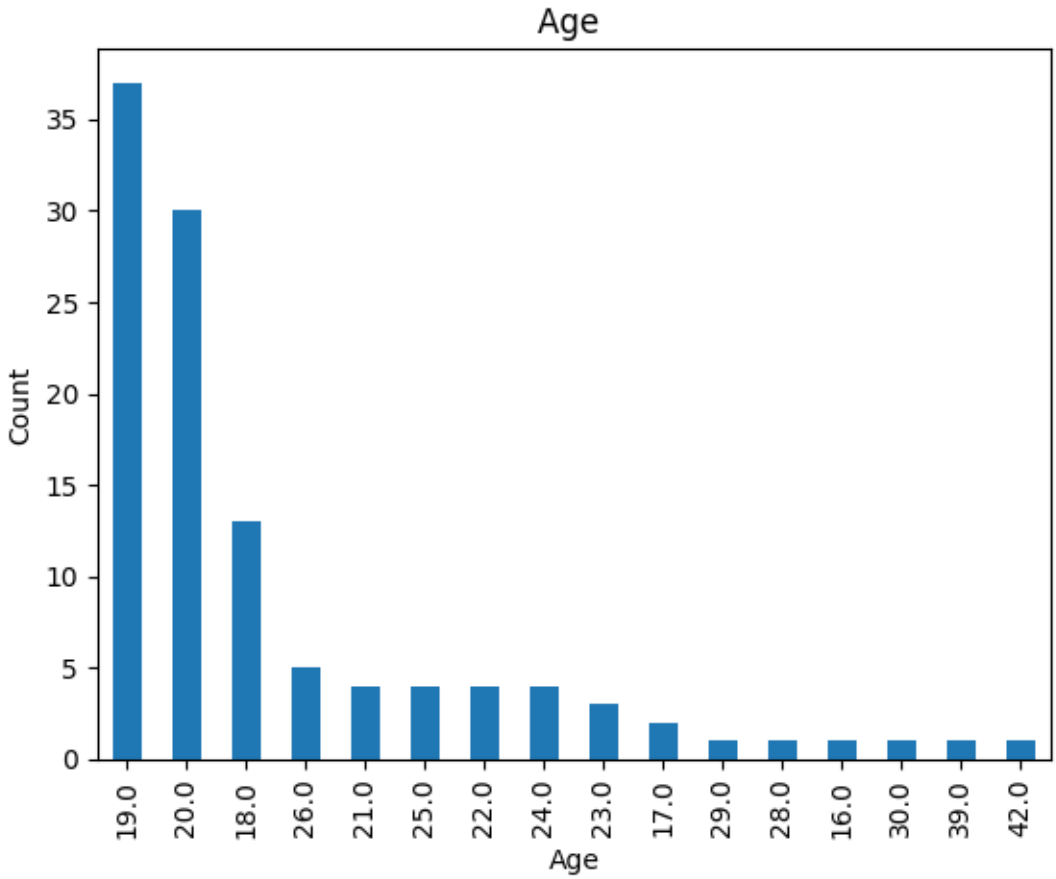

Figure 2 shows the age distribution of participants. Most respondents fall within the 18-20 age range, with fewer responses from older groups. This suggests that the dataset is concentrated around typical university-aged individuals, aligning with the population of interest.

Figure 3. Sleep Hours Distribution

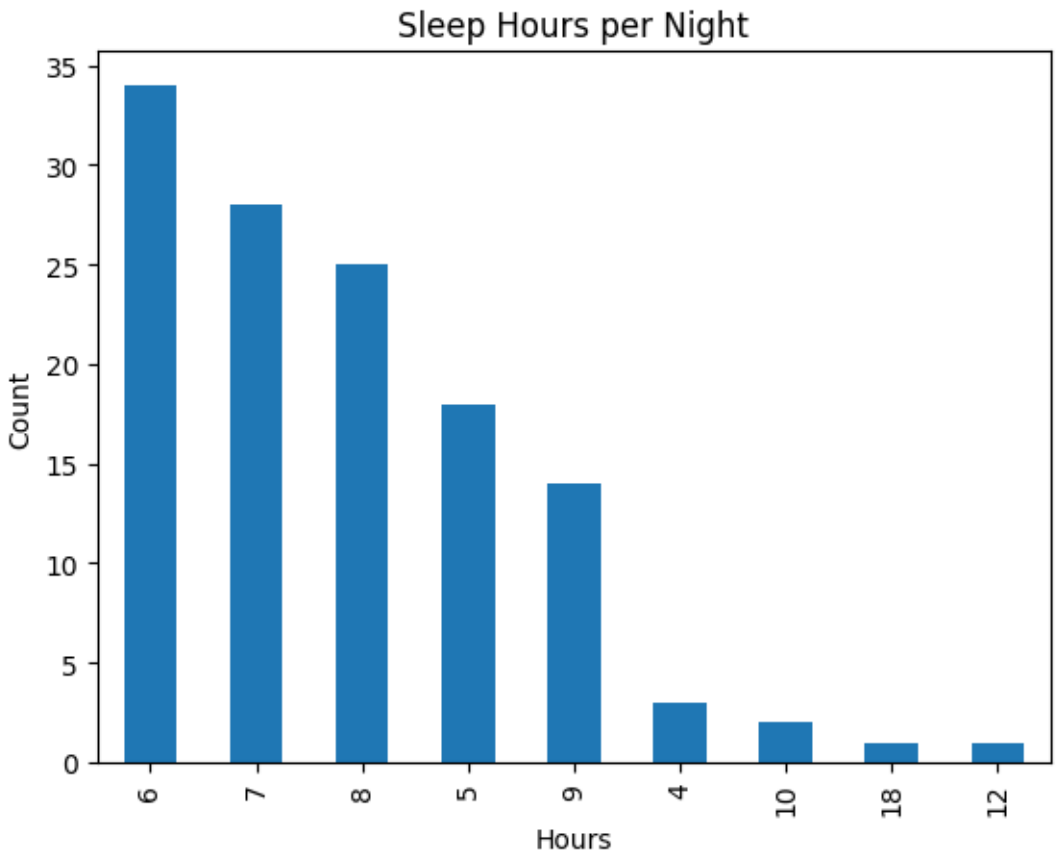

Figure 3 shows the distribution of sleep hours among participants. The most common sleep duration is 6 hours, followed by 7 and 8 hours. This indicates that many participants are getting slightly less than the recommended 7-9 hours of sleep. The responses are clustered within the moderate range, with fewer extreme values.

In addition to the visual distribution, descriptive statistics were calculated for sleep duration. The mean sleep duration is approximately 6.97 hours, while the median is 7 hours, indicating that half of the participants sleep around 7 hours per night. The mode is 6 hours, which represents the most frequently reported sleep duration among respondents. These values suggest that most participants cluster between 6 and 7 hours of sleep, slightly below the recommended range of 7-9 hours.

Figure 4. Sleep Quality

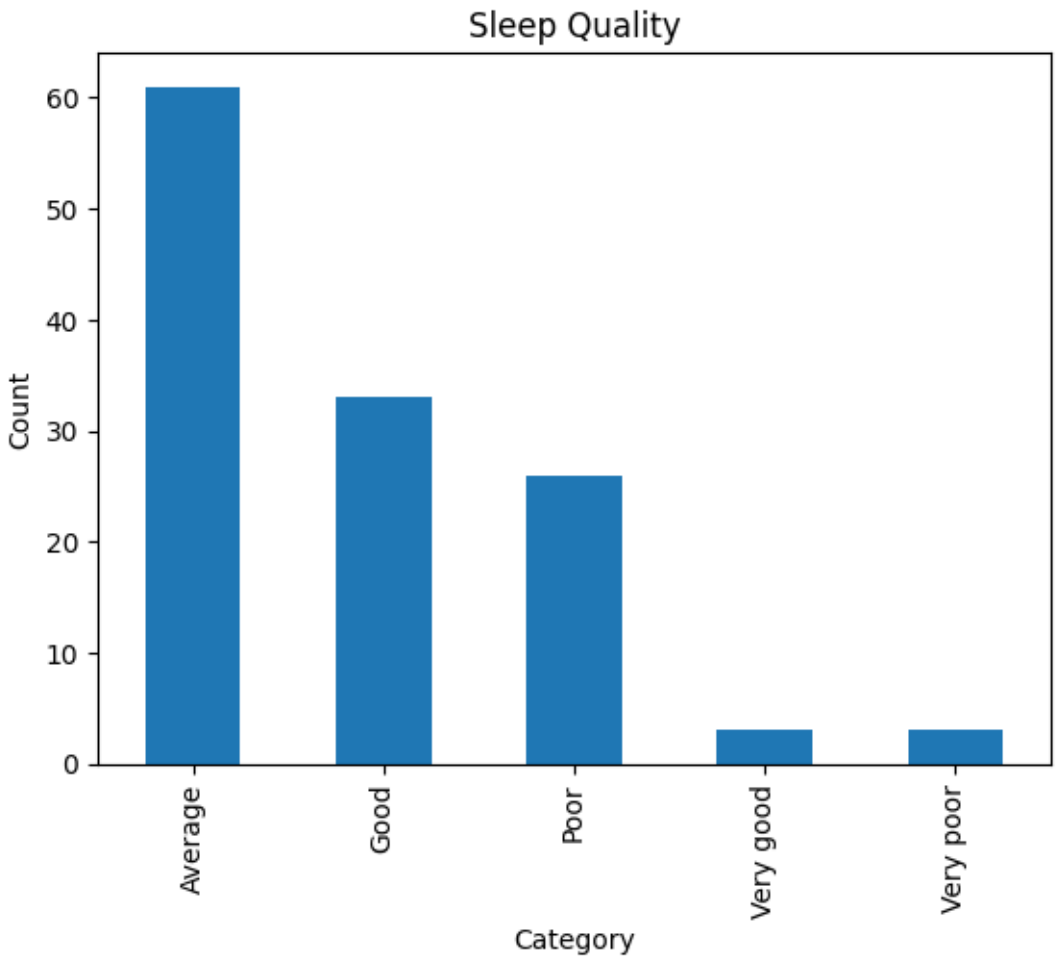

Figure 4 shows the distribution of sleep quality ratings among participants. Most respondents rated their sleep as average, followed by good, while fewer reported poor sleep quality. This suggests that sleep quality is generally moderate, with limited extreme cases.

Figure 5. Caffeine Consumption per Day

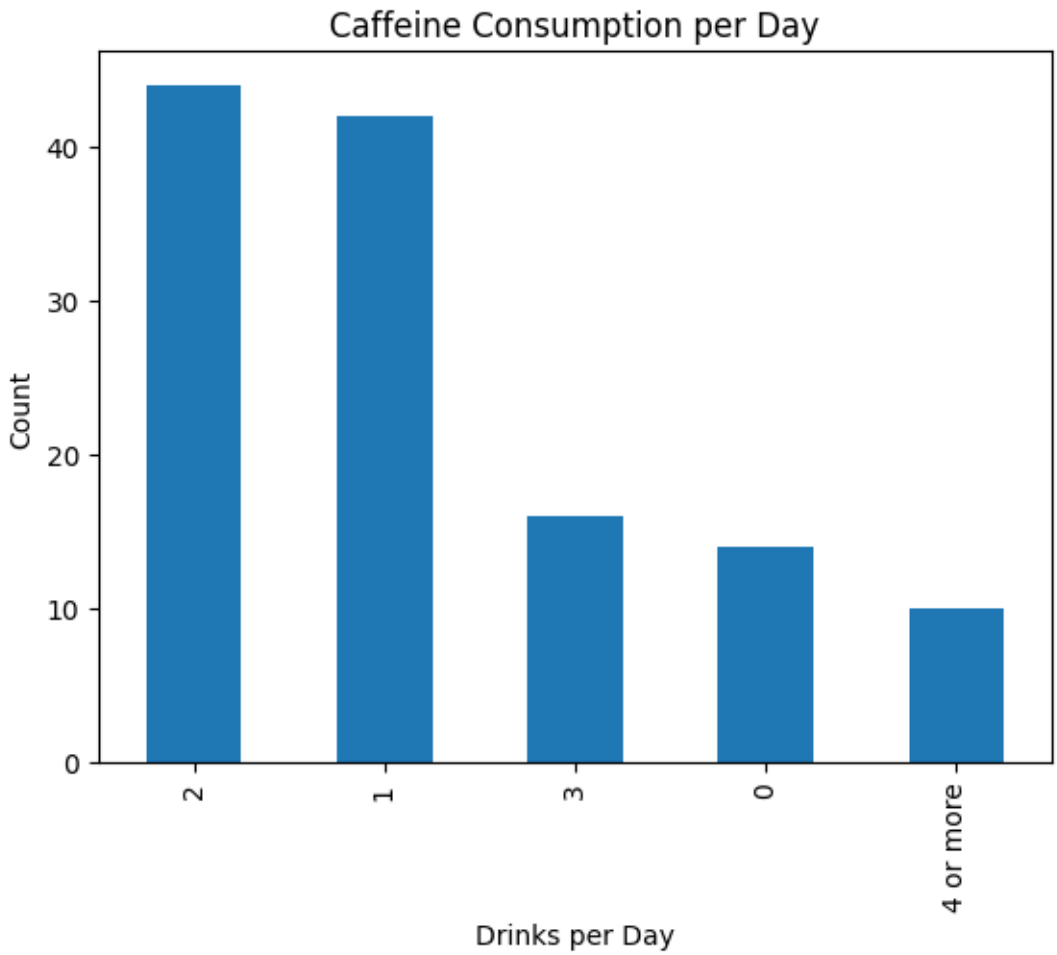

Figure 5 shows the distribution of daily caffeine consumption. The most common level is 2 drinks per day, followed by 1 and 3 drinks, while fewer participants consume 0 or more than 4 drinks. This indicates that moderate caffeine intake is the dominant pattern among respondents.

Descriptive statistics for caffeine consumption show that the mean intake is approximately 1.73 drinks per day, while the median is 2 drinks, indicating that a typical participant consumes around two caffeinated drinks daily. The mode is also 2 drinks, meaning it is the most common response among participants. This reinforces the observation that moderate caffeine consumption is the dominant pattern in the dataset.

Figure 6. Time of Caffeine Consumption

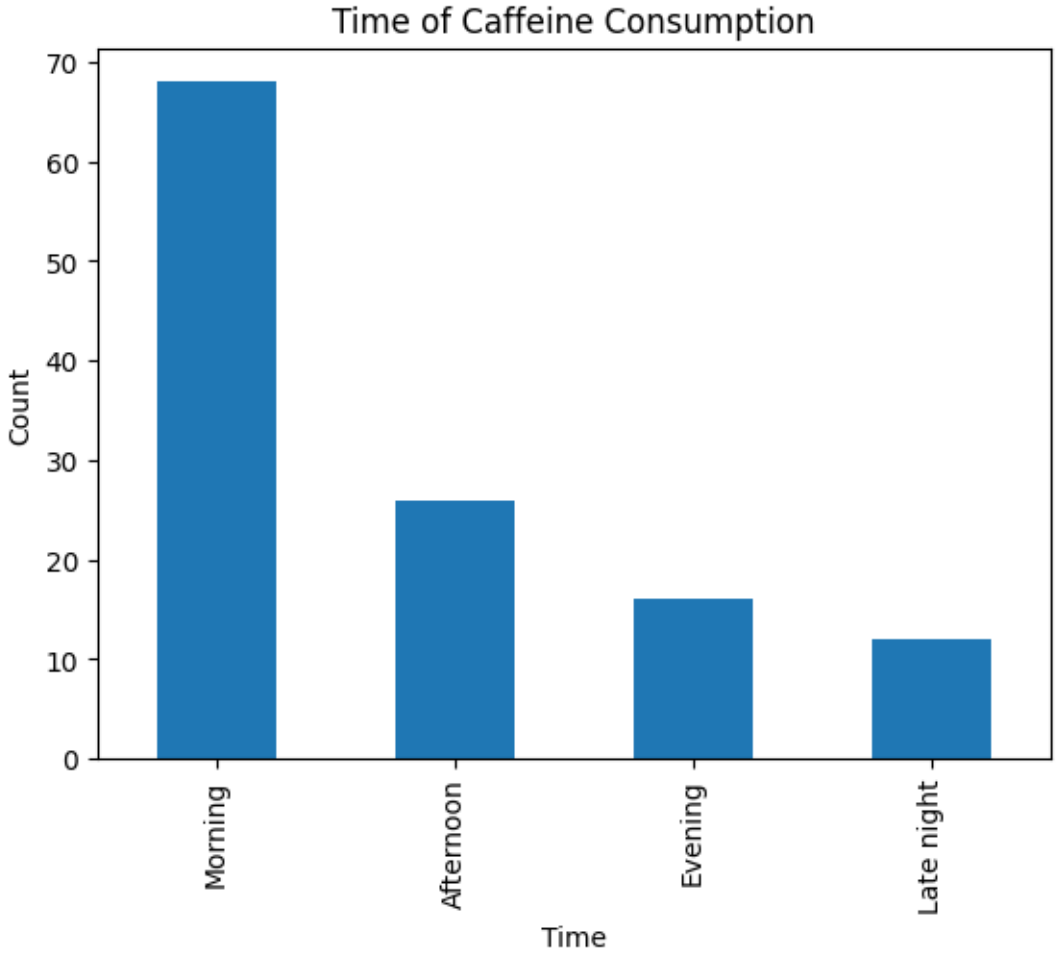

Figure 6 shows the time of caffeine consumption among participants. Most individuals consume caffeine in the morning, followed by the afternoon, evening and late night. This suggests that caffeine is primarily used for energy earlier in the day, although some participants consume it later, which may affect sleep.

Figure 7. Caffeine Before Sleep

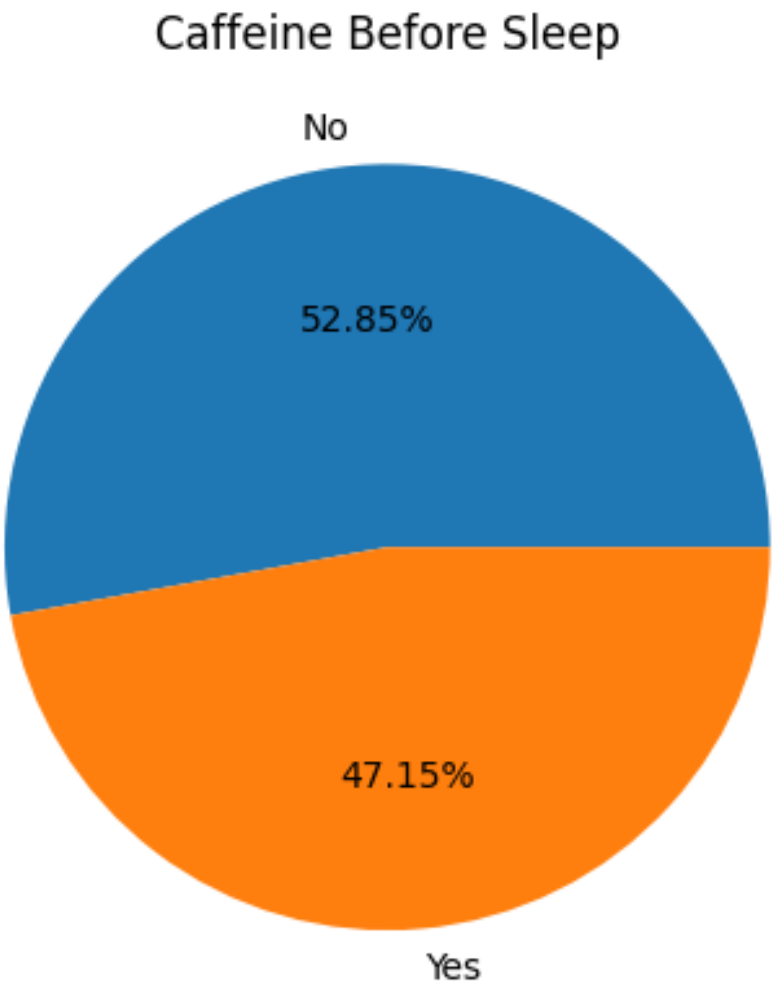

Figure 7 shows whether participants consume caffeine withing 4 hours before sleep. Approximately 52.84% of respondents do not consume caffeine before sleep, while 47.15% do. This nearly equal split indicates that a significant portion of individuals engage in late caffeine consumption, which could negatively impact their sleep. The high percentage of “yes” responses suggest that this behaviour is common and may contribute to reduced sleep quality or duration.

Figure 8. Does Caffeine Affect Sleep?

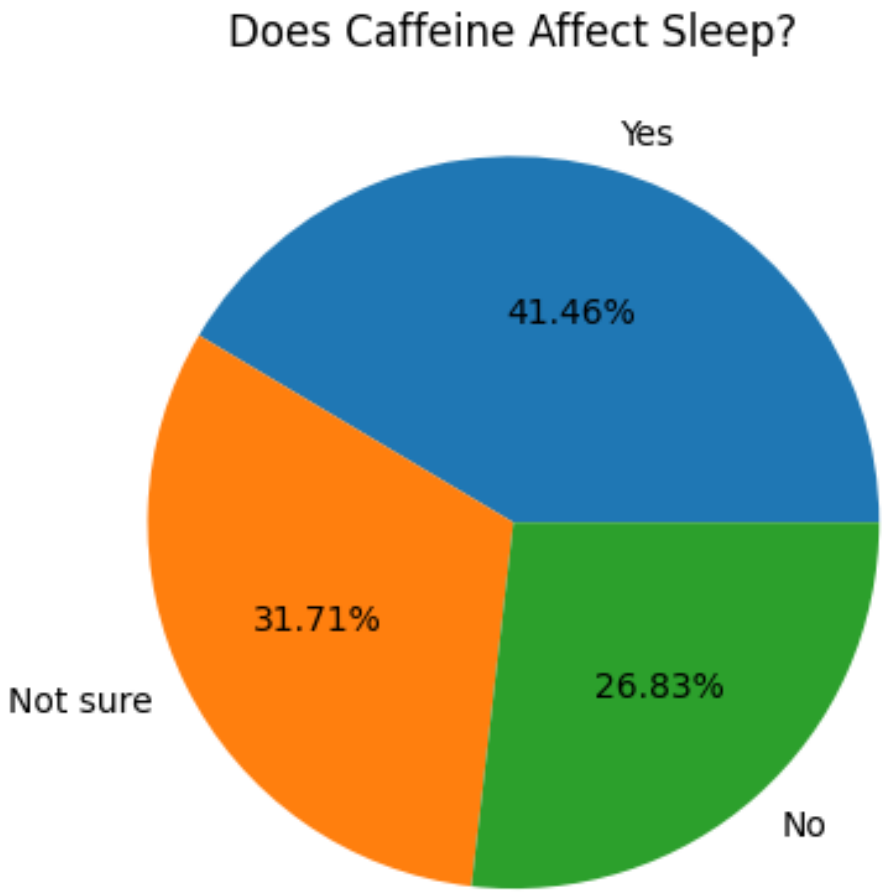

Figure 8 shows participants' perceptions of whether caffeine affects their sleep. Around 41.46% of participants believe caffeine affects their sleep, 31.71% are unsure and 26.83% believe it does not.

This indicates that although a large portion of respondents recognize the impact of caffeine, there is still considerable uncertainty. The variation in responses suggests that the effects of caffeine may differ between individuals, possibly due to tolerance levels or personal habits.

Figure 9. Caffeine vs Sleep Hours

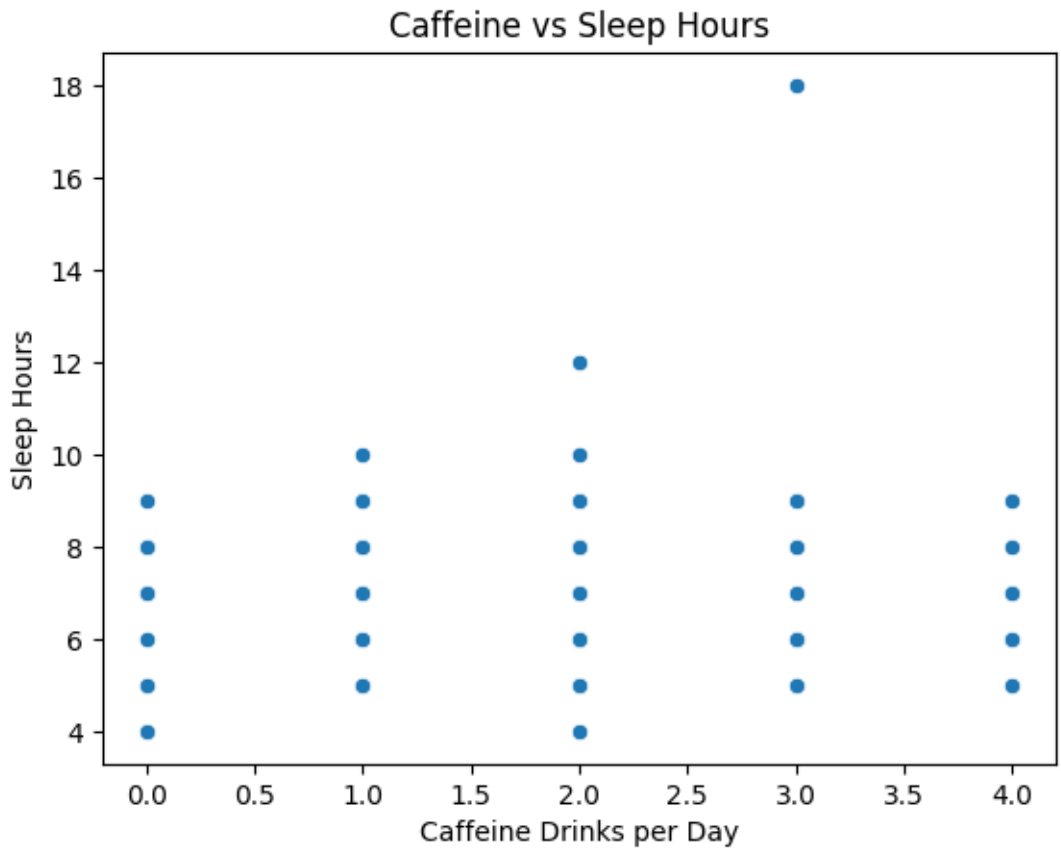

Figure 9 shows the relationship between caffeine consumption and sleep duration. The data suggests a general trend where individuals who consume higher amounts of caffeine tend to sleep fewer hours.

However, this trend is not perfectly consistent across all participants, indicating that caffeine is not the only factor influencing sleep duration. Other variables such as stress, daily routine and screen time may also play a role in how much sleep individuals get. The relationship appears to be weak to moderate rather than strong, suggesting other factors also influence sleep duration.

Conclusion

The results of this study suggest a moderate relationship between caffeine consumption and sleep patterns. While higher caffeine intake appears to be associated with reduced sleep duration and lower sleep quality, the relationship is not absolute.

This indicates that caffeine is likely one of the several factors influencing sleep, rather than the sole determinant. Therefore, the hypotheses is partially supported by the data but cannot be fully confirmed.

Any potential issues

One limitation of this study is the use of convenience sampling, which may not fully represent the wider population. Additionally, self-reported data may introduce inaccuracies due to memory errors or subjective interpretation of sleep quality.

Another limitation is that the study does not account for other influencing factors such as stress levels, screen time, or physical activity.